

INTRODUCTION

The second law of thermodynamics states that the entropy of an isolated system must increase over time. Entropy is commonly interpreted as a measure of disorder, and in most physical systems, this corresponds to particles spreading out uniformly in space. For example, a gas confined to a small region will naturally expand to fill the entire volume available to it, increasing entropy.

However, gravitational systems appear to behave in a completely opposite way. When particles interact through gravity, they do not spread out instead, they attract each other and form dense clusters. This leads to the formation of structures such as stars, galaxies, and clusters of galaxies. From a spatial perspective, this looks like a transition from disorder to order, which seems to contradict the second law.

This creates a fundamental paradox:

If entropy must increase, why do gravitational systems become more ordered in space?

The resolution of this paradox requires a deeper understanding of entropy. Entropy is not simply a measure of spatial distribution, but rather a measure of the number of accessible microstates in phase space, which includes both position and velocity.

In this project, this paradox is investigated numerically by simulating a self-gravitating system of particles. The system evolves according to Newtonian gravity, and three different entropy measures are tracked simultaneously:

- Spatial entropy (based on position distribution)
- Velocity entropy (based on speed distribution)
- Phase-space entropy (combined measure)

By comparing these three quantities, the project aims to determine how entropy evolves in a gravitational system and whether the second law remains valid.

THEORETICAL FRAMEWORK

PART I: STATISTICAL MECHANICS

1. Statistical Description of Many-Particle Systems

In systems containing a large number of interacting particles, tracking the motion of each particle individually becomes impractical. Instead, statistical mechanics describes the system in terms of collective behavior using probability distributions rather than individual trajectories.

A distinction is made between microstates, which represent the exact positions and velocities of all particles, and macrostates, which correspond to observable properties such as energy or density. Entropy quantifies the number of microstates associated with a given macrostate.

2. Phase Space Representation

The complete state of the system is represented in phase space, which combines position and velocity coordinates. Instead of following individual particles, the system is described using a distribution function $f(r, v, t)$, which represents the density of particles in phase space. This function gives the density of particles in phase space at a given time. It contains all the statistical information about the system.

3. Entropy in Discrete Systems

In numerical simulations, entropy is evaluated using discrete probability distributions. The entropy is defined as:

$$S = -\sum p_i \log p_i$$

where p_i represents the probability of particles within each region of space or velocity. This form allows entropy to be computed directly from simulation data.

4. The Second Law and Gravitational Systems

The second law of thermodynamics states that the entropy of an isolated system does not decrease over time. However, in self-gravitating systems, particles tend to cluster due to mutual attraction, leading to increased spatial order.

This creates an apparent contradiction: while thermodynamics predicts increasing disorder, gravitational systems evolve toward structured configurations. The key

question addressed in this project is whether this behavior represents a true decrease in entropy or a redistribution of entropy within the system.

PART II: GRAVITATIONAL PHYSICS

1. Newtonian Gravity and the N-Body Problem

In a system of interacting particles, each particle experiences a gravitational force due to all other particles. The total force on a particle is obtained by summing the contributions from every other particle in the system.

This defines an N-body system in which all particles are mutually coupled, and the motion of each particle depends on the entire system. As the number of particles increases, the number of interactions grows rapidly, making analytical solutions impractical. The system therefore exhibits complex and nonlinear collective behavior, requiring numerical methods for its evolution.

2. The Gravitational Softening Parameter

In gravitational systems, forces increase sharply as the distance between particles becomes very small. This can lead to numerical instabilities and unphysically large accelerations.

To avoid this, the effective separation between particles is modified as:

$$r_{ij} \rightarrow \sqrt{r_{ij}^2 + \epsilon^2}$$

where **ϵ is the softening parameter**. This modification ensures that the force remains finite at short distances while having negligible effect at large separations. As a result, it stabilizes the simulation and prevents artificial divergences during close particle interactions.

3. Numerical Integration — Velocity Verlet Method

The time evolution of the system is computed using the Velocity Verlet integration scheme, which provides improved stability and better energy conservation compared to simpler methods.

The equations are:

$$\begin{aligned}\vec{v}(t + \frac{dt}{2}) &= \vec{v}(t) + \frac{1}{2}\vec{a}(t) dt \\ \vec{r}(t + dt) &= \vec{r}(t) + \vec{v}(t + \frac{dt}{2}) dt \\ \vec{v}(t + dt) &= \vec{v}(t + \frac{dt}{2}) + \frac{1}{2}\vec{a}(t + dt) dt\end{aligned}$$

This method updates positions and velocities in a staggered manner, reducing numerical errors and ensuring that physical quantities such as total energy remain approximately

conserved over time. Despite these limitations, the method effectively captures the overall dynamical behavior of the system.

4. Gravitational Collapse and Dynamical Stabilization

When particles are initially distributed with low velocities, gravitational attraction dominates their motion, causing them to move toward regions of higher density. This leads to gravitational collapse, where the system contracts and forms a dense central structure.

During this process, gravitational potential energy decreases while kinetic energy increases, reflecting the conversion of potential energy into motion. As particles interact, energy is redistributed among them, leading to a wide range of particle velocities.

The collapse does not continue indefinitely. As particle velocities increase, they begin to counteract further contraction. The system eventually reaches a dynamically stable state in which gravitational attraction is balanced by particle motion. In this state, the overall structure remains approximately constant, even though particles continue to move.

PART III: ENTROPY AND PHASE-SPACE ANALYSIS

1. Phase-Space Evolution and Liouville's Principle

A many-particle system is fully described when both positions and velocities are considered together. This combined representation is known as phase space. As the system evolves, the distribution of particles changes shape within this space.

However, this evolution follows a fundamental constraint: the density of the system in phase space remains constant along its motion, expressed as:

$$\frac{df}{dt} = 0$$

This implies that the phase-space distribution behaves like an incompressible flow. It can stretch and fold into complex structures, but its total volume remains unchanged. As a result, even though particles cluster in physical space, the underlying phase-space distribution is preserved.

This has an important implication for entropy. The apparent increase in order observed in position space does not correspond to a true decrease in entropy, but rather reflects a redistribution within phase space.

2. Entropy Measurement and Coarse Representation

In numerical simulations, entropy is computed using discrete probability distributions rather than continuous functions. The entropy is defined as:

$$S = -\sum p_i \log p_i$$

where p_i represents the probability of particles within each region.

Since the system is divided into finite bins, the resolution is limited. As the phase-space distribution evolves, it develops fine structures that cannot be fully resolved at this scale. Consequently, some information about the system is lost.

The computed entropy therefore represents a **coarse-grained** description of the system. This explains why entropy appears to change during evolution, even though the underlying phase-space distribution remains conserved.

3. Energy Redistribution and Velocity Evolution

As the system evolves under gravitational interaction, energy is continuously redistributed among particles. During collapse, gravitational potential energy decreases while kinetic energy increases, leading to a broader range of particle velocities. Initially, particle velocities are similar, resulting in a narrow distribution. Over time, interactions

cause some particles to gain significant kinetic energy while others remain slower, producing a wide velocity distribution. This increase in velocity diversity corresponds to an increase in velocity entropy.

This rapid redistribution of energy during collapse is referred to as **violent relaxation**. The rapidly changing gravitational potential leads to strong interactions, allowing energy to be redistributed efficiently across the system. As a result, some particles become tightly bound, while others gain enough energy to move outward.

4. Entropy Redistribution in the System

The apparent contradiction in gravitational systems arises when only spatial distribution is considered. As particles cluster, spatial entropy decreases, suggesting increased order. However, at the same time, the velocity distribution becomes more diverse, leading to an increase in velocity entropy.

When both position and velocity are considered together, it becomes clear that the system is not losing entropy, but redistributing it between spatial and velocity components. The decrease in spatial entropy is compensated by the increase in velocity entropy. This demonstrates that the second law of thermodynamics is not violated.

SIMULATION FRAMEWORK AND METHODOLOGY

1. Integration of Theory with Simulation

The simulation applies concepts from gravitational physics and statistical mechanics to study the evolution of a self-gravitating system. Gravitational interaction governs the motion of particles, leading to collapse and clustering, while entropy is used to quantify the distribution of the system in both position and velocity space. The simulation therefore provides a numerical realization of the theoretical framework.

2. System Initialization

The system consists of a set of particles distributed randomly in two-dimensional space with small initial velocities. This represents an initially diffuse configuration with no preferred structure. The mean velocity of the system is adjusted to zero to ensure that there is no overall drift of the system.

3. Gravitational Interaction

Each particle interacts with all others through gravitational forces, forming a coupled N-body system. The total force on each particle is obtained by summing contributions from all other particles. To prevent numerical instability at very small separations, a softening parameter is introduced, ensuring that the force remains finite and physically meaningful throughout the simulation.

4. Time Evolution (Velocity Verlet Method)

The system is evolved over time using the Velocity Verlet integration scheme. In this method, particle velocities and positions are updated using forces evaluated at successive time steps. This approach provides stable numerical evolution and helps maintain approximate conservation of total energy.

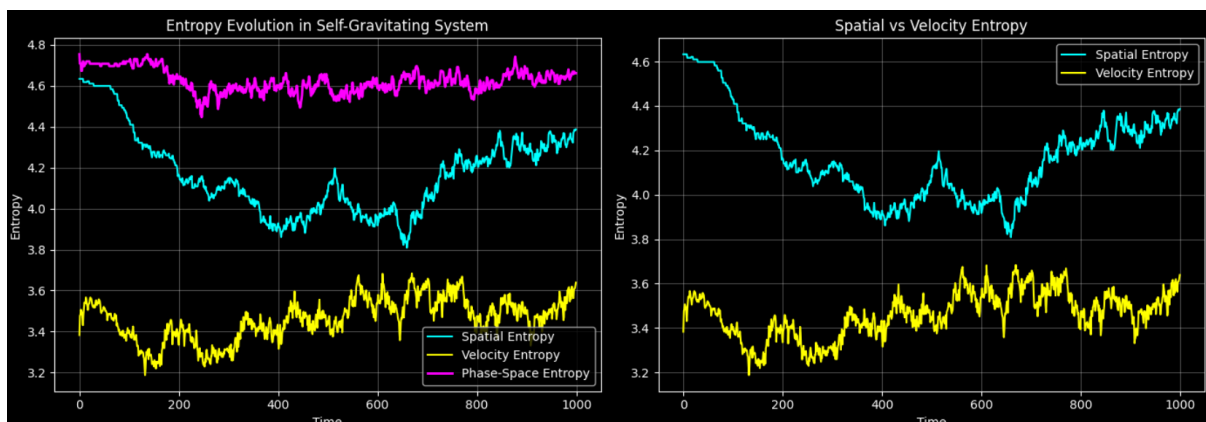
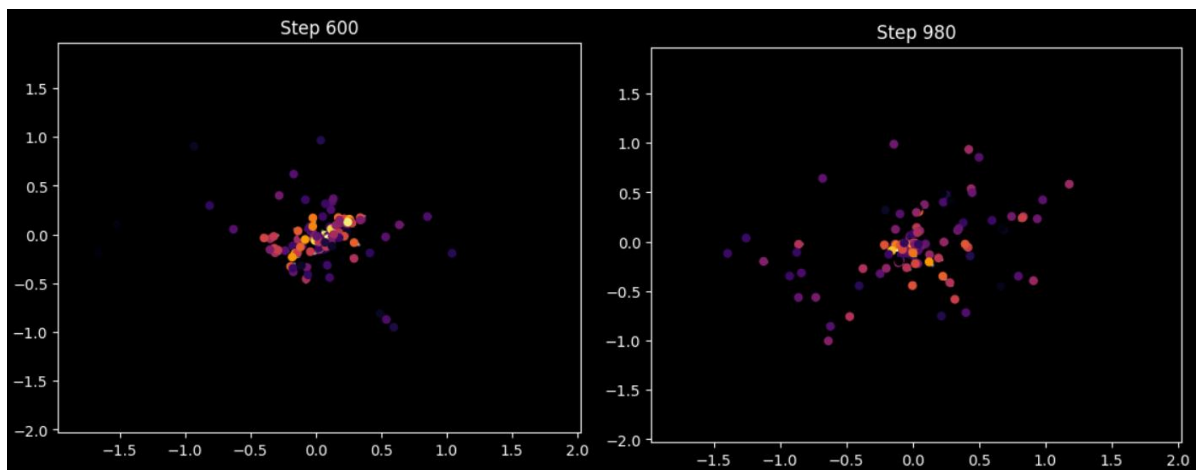
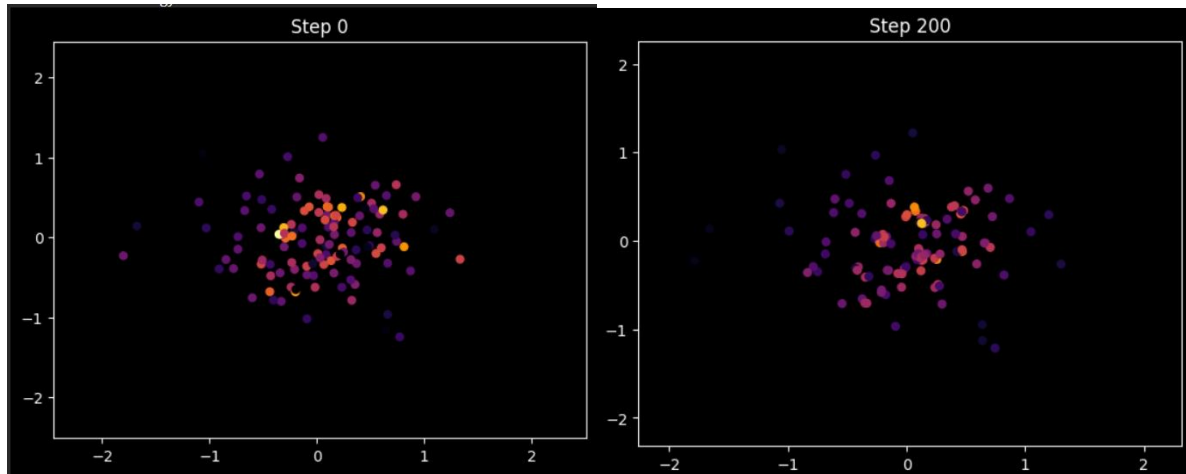
5. Physical Quantities Calculation

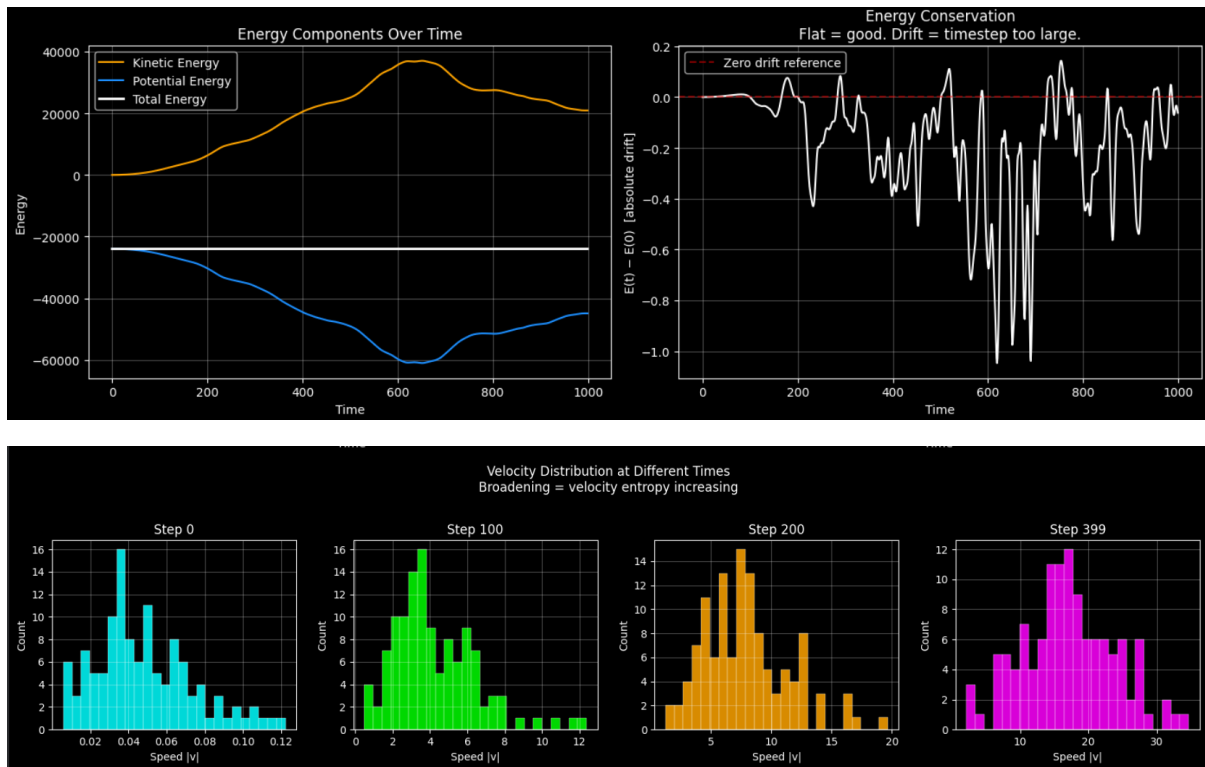
At each time step, key physical quantities are computed to analyze the system:

- **Entropy:** Calculated using discrete probability distributions obtained by dividing the system into finite bins. Spatial entropy measures particle distribution in space, while velocity entropy measures the spread of particle speeds. Phase-space entropy combines both aspects.

- **Energy:** Kinetic, potential, and total energy are evaluated to study energy redistribution and verify conservation throughout the simulation.

RESULTS AND DISCUSSION





1. Particle Evolution (Scatter Plots):

The system evolves from an initially dispersed distribution into a dense central cluster due to gravitational attraction. At later stages, some particles move outward, indicating energy redistribution and dynamic interactions within the system.

2. Entropy Evolution:

Spatial entropy decreases as particles cluster, while velocity entropy increases due to the spread in particle speeds. Phase-space entropy remains relatively stable, showing that the system redistributes entropy rather than violating thermodynamic principles.

3. Energy Evolution:

During collapse, potential energy decreases while kinetic energy increases, reflecting the conversion of gravitational energy into motion. The total energy remains approximately constant, indicating stable numerical integration.

4. Energy Conservation (Drift Plot):

The total energy shows small fluctuations over time, but no significant drift, confirming that the simulation maintains overall energy conservation.

5. Velocity Distribution:

The velocity distribution broadens over time, transitioning from a narrow initial distribution to a wider spread. This indicates increasing velocity dispersion and supports the rise in velocity entropy.

LIMITATIONS

The simulation represents an idealized model of a self-gravitating system and can be further refined for greater physical accuracy. Increasing the number of particles would improve resolution and allow better representation of large-scale structure formation. The use of a softening parameter ensures numerical stability, though more advanced techniques could model short-range interactions more precisely. Additionally, entropy is evaluated using discrete bins, and higher-resolution methods could capture finer phase-space structures. While the numerical integration conserves energy well overall, smaller time steps or adaptive methods could further enhance accuracy.

CONCLUSION

This project presents a numerical study of a self-gravitating system using a particle-based simulation. The results show that gravitational attraction leads to the formation of dense clusters, while simultaneous energy redistribution produces a broader range of particle velocities. This demonstrates that although the system appears more ordered in space, entropy is not reduced but redistributed between position and velocity.

The consistent behavior of entropy trends and the approximate conservation of total energy confirm that the simulation successfully captures the essential physics of gravitational many-body systems. Overall, the study provides a clear and coherent demonstration of how gravitational dynamics and statistical mechanics are fundamentally connected.

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